

Bi Variate Analysis

Hamidullah, Rezae
ITC 255
Spring Semester 2021

1. Introduction:

We use the data collected from 194 individuals of a company to study a number of associations between the variables of interest.

We have collected data on variables. Our population of interest consists of all employees of a tech company in Kabul. We employ bi variate analysis techniques to statistically study the following questions:

- [1] Are age and gender related in our population?
- [2] Are marital status and gender related in our population?
- [3] Are age and education are related in our population?

2. Methodology:

2.1. Population and data:

Our population consists of all 194 individuals of a tech company in Kabul. The company offers different technology related services, such as application development, networking, and database design. The data on the variables of interest were collected from all employees of the company online via Google form and then cleaned in Excel. R and Excel are used to analyze the data.

2.2. Analysis:

The collected data are analyzed using both uni- and bi variate methods. They include distribution table, graphical methods, and numerical techniques. Specifically we use, uni variate numerical methods to present the summary statistics of the variables. Furthermore, we employ chi-square, t-stat, and correlation as well as joint bar graph and scatter plot to study different associations.

3. Results:

3.1. Summary statistics:

In this section we discuss the population characteristics. Table 1 presents the summary statistics of the quantitative variables.

Table 1. Summary statistics of the quantitative variables

<i>Variable</i>	<i>Min</i>	<i>Mean</i>	<i>Median</i>	<i>St.Dev</i>	<i>3^d Quantile</i>	<i>Max</i>
Age	54.00	60.48	60.00	4.208586	64.00	73.00
Education	8.00	13.57	12.00	2.578799	16.00	18.00

Furthermore, we use graphical representation methods to summarize the qualitative variables. Figure 1 shows the Gender distribution.

Distribution of Gender

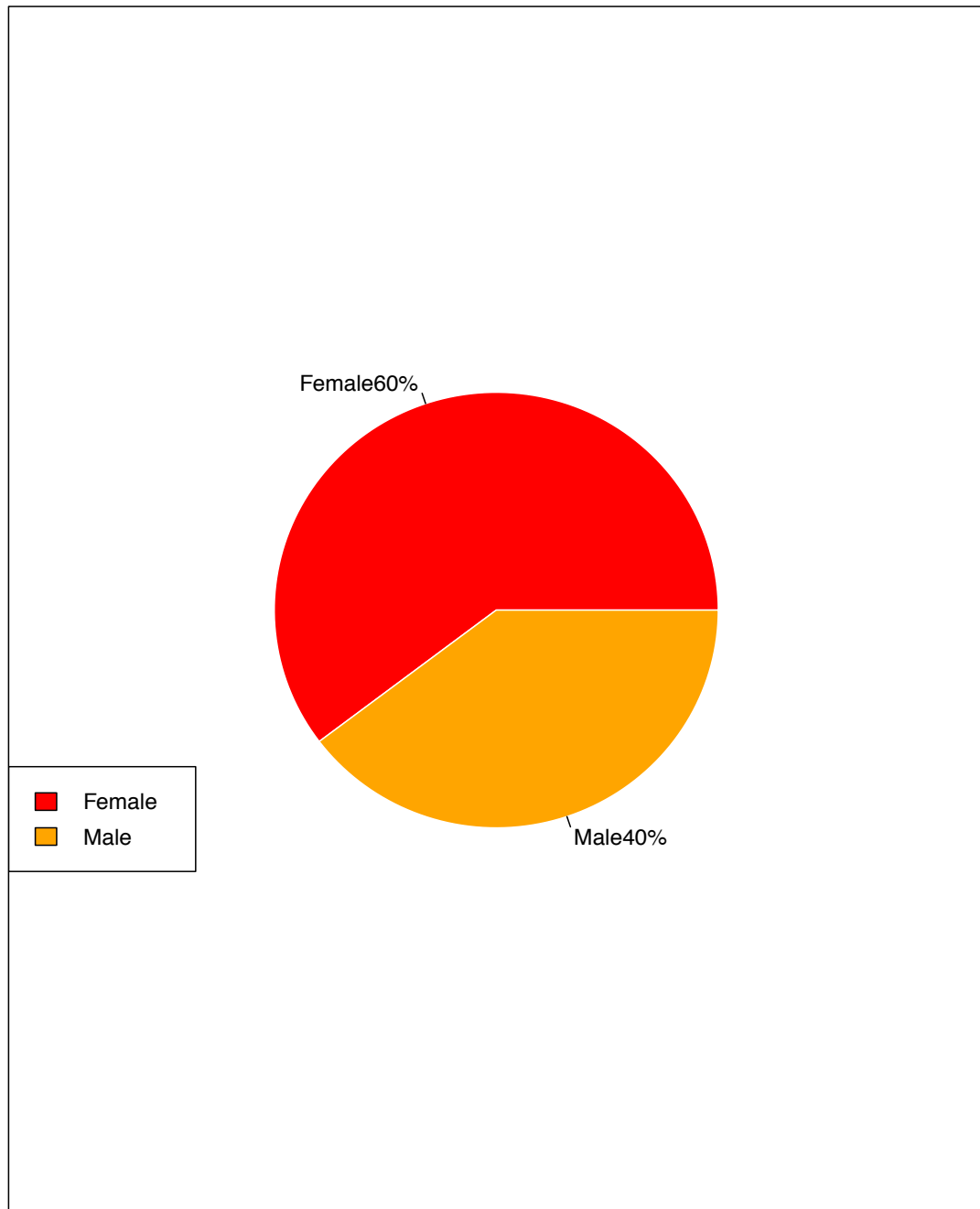
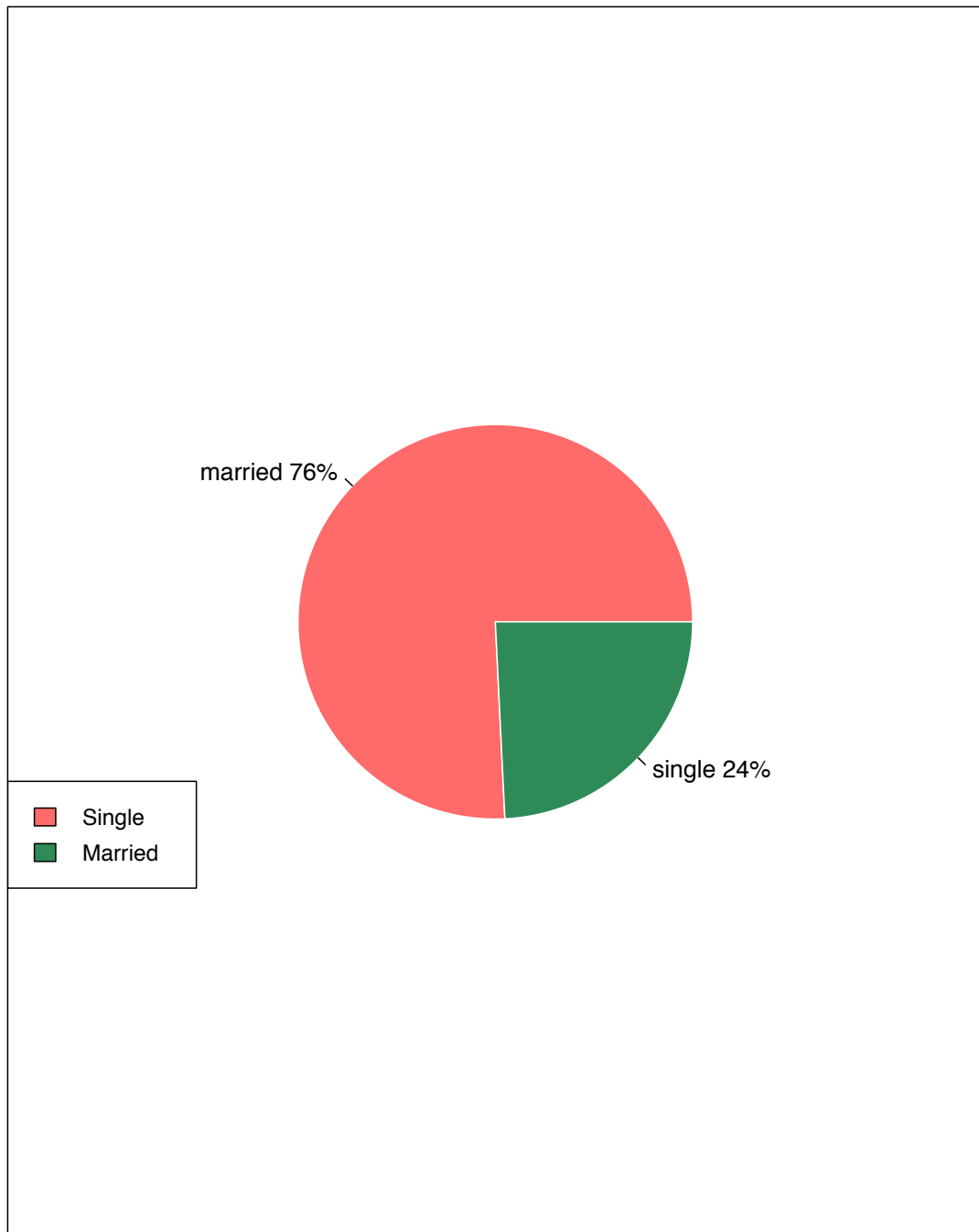


Figure 1. Distribution of Gender

As we can see we have 60% of female participants who decided to save their retirement in later years of life, but 40% of participants who decided to save their retirements were male.

Distribution of Martial Status



As we can see we can assert that married people are more likely to decide to save their retirement savings in their later years in life. 76% of people who decided to save their retirements were married and 24% of those who saved their retirements were single.

3.2. Bi variate Analysis

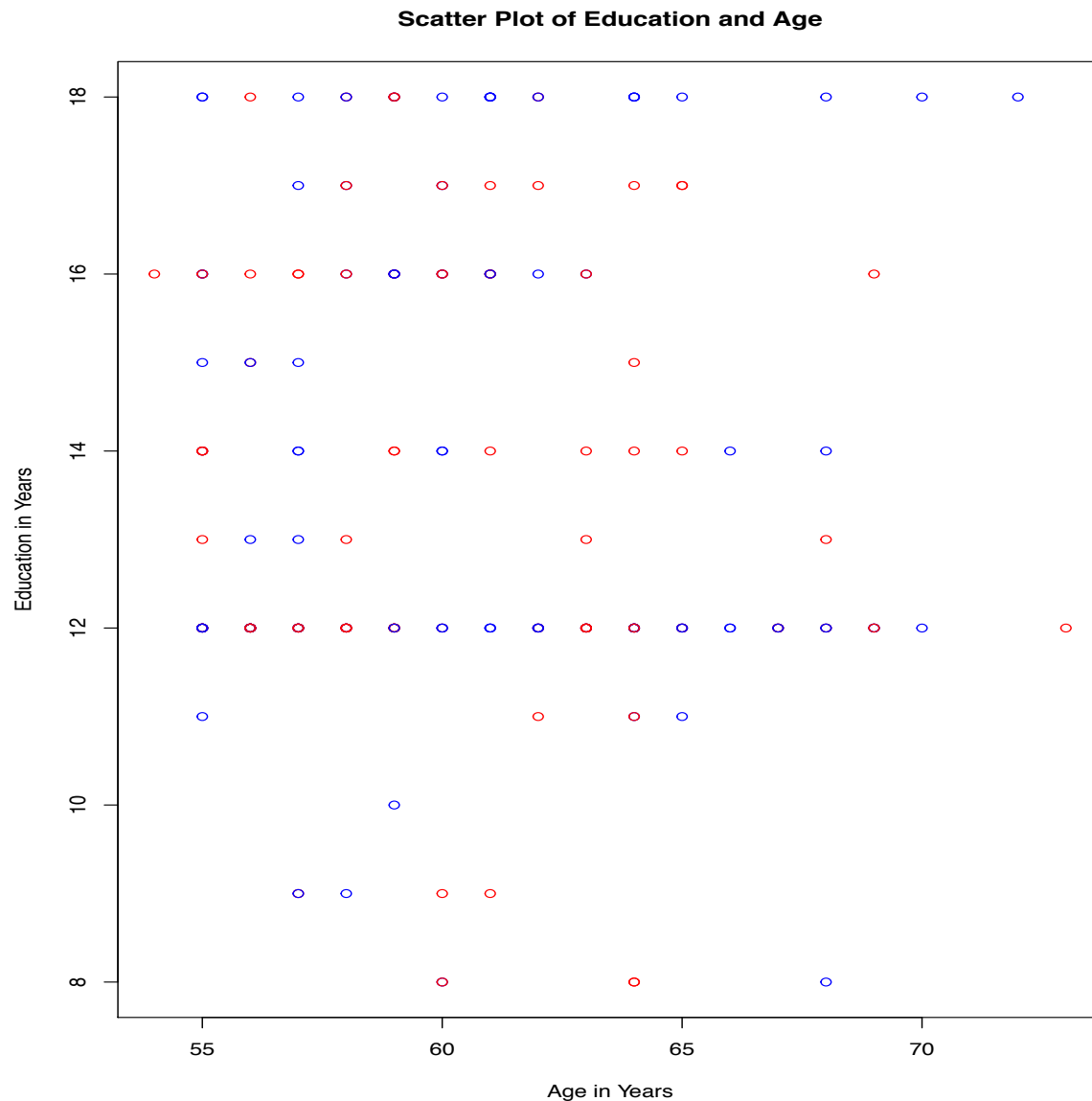
This section discusses the association of the concerned variable by applying different bi variate analysis.

The Chi-Square statistic of Gender and Martial Status is (38.662) which show a significant association between the two variables. This means that Gender of retired people are very closely related to their Martial Status when it comes making decisions about their retirement savings.

A t-test of Gender and age is ($t = -4.9505$) which shows a strong association between the age and Gender of participants in deciding about their retirement savings. This means that gender of participants affects their decisions in saving for retirement in later years of their lives.

Figure 2 presents the scatter plot of Education and age.

3.3. Linear Regression



To

Figure 2: Scatter plot of Education and age

Figure 2 suggest a weak linear association between the two variables. To measure this numerically we compute their correlation which is (-0.04095366) showing a strong indirect relation. This means that there is a very weak association between age and Education of individuals whose financial decisions in retirement savings is being studied. People with higher age have less years of educational experience, and people with lower age are having higher educational experiences.

Appendix:

Part 1 here:

#Question 01:

```
> install.packages("wooldridge")
> library("wooldridge")
> data("pension")
> p=pension
> View(p)
> help(p)
> #Part one:
> #Frequency Distribution Table of Quant. Variable (age)
> Quant.function= function(x){
+   cl=3
+   classlimit=seq(min(x),max(x), by = cl)
+   br=cut(x,classlimit,right = T,include.lowest = T)
+   t=table(br)
+   relative.freq=round(prop.table(t),2)
+   Cumulative.freq=cumsum(relative.freq)
+   final.table=cbind(t, relative.freq,Cumulative.freq)
+   return(final.table)
+ }
> Quant.function(p$age)
      t relative.freq Cumulative.freq
[54,57] 58      0.30      0.30
(57,60] 49      0.25      0.55
(60,63] 36      0.19      0.74
(63,66] 31      0.16      0.90
(66,69] 16      0.08      0.98
(69,72]  3      0.02      1.00
##Freq. Distribution table for Quant. Variable (educ):
> Quant.function= function(x){
+   cl=3
+   classlimit=seq(min(x),max(x), by = cl)
+   br=cut(x,classlimit,right = T,include.lowest = T)
+   t=table(br)
+   relative.freq=round(prop.table(t),2)
+   Cumulative.freq=cumsum(relative.freq)
+   final.table=cbind(t, relative.freq,Cumulative.freq)
+   return(final.table)
+ }
> Quant.function(p$educ)
      t relative.freq Cumulative.freq
```



```

[8,11] 16      0.09      0.09
(11,14] 114     0.67      0.76
(14,17] 40      0.24      1.00
> #Part two: frequency distribution table of Qual. variable married (Marital Status)
> #First we need to substitute some variables with understandable terms
> help(p)
> p$married=gsub("1", "married", p$married)
> p$married=gsub("0", "single", p$married)
> colnames(p)[6]="Marital Status"
> #Making sure that the variable of interest has been changed and viewing the Frequency
Distribution Table
> View(qual.fun(p$`Marital Status`))
> #Part two: frequency distribution table of Qual. variable female (Gender)
> #First we need to substitute some variables with understandable terms
> help(p)
> p$female=gsub("1", "Female", p$female)
> p$female=gsub("0", "Male", p$female)
> colnames(p)[5]="Gender"
> #Making sure that the variable of interest has been changed
> View(qual.fun(p$`Gender`))
> d=(qual.fun(a$Gender))
> d
      Abs.freq re.freq cum.freq
Female   117   0.6   0.6
Male     77   0.4   1.0

```

Part 2 here:

##Question 02:

##1a. Pie chart for Qual. Variable (Marital Status)

```

> f=qual.fun(p$`Marital Status`)
> f
      Abs.freq re.freq cum.freq
married   147   0.76   0.76
single    47   0.24   1.00
> pie(f$Abs.freq, labels = paste(cl, round(prop.table(f$Abs.freq)*100), "%", sep = ""),
+   cex=1.1,
+   col=c("indianred1","seagreen"),
+   border = "white",
+   radius = 0.8,
+   main = "Distribution of Marital Status")
>
>
> legend("bottomleft", legend = c("Single", "Married"), fill =c("indianred1","seagreen"))
>

```

```

> box(which = "plot", lty = 1)
##2a. Pie Chart of Qual. Variable Gender
> Cl=c("Female", "Male")
> cl=paste(Cl, d$re.freq*100)
> g=qual.fun(p$Gender)
> g
      Abs.freq re.freq cum.freq
Female   117    0.6    0.6
Male     77    0.4    1.0
> pie(g$Abs.freq, labels = paste(cl, round(prop.table(g$Abs.freq)*100), "%", sep = ""),
+   cex=1,
+   col=c("red", "orange"),
+   border = "white",
+   radius = 0.9,
+   main = "Distribution of Gender")
>
>
> legend("bottomleft", legend = c("Female", "Male"), fill = c("red", "orange"))
>
> box(which = "plot", lty = 1)
> #1b. Graph of Quant. Variable (age)
> age=Quant.function(p$age)
> h=hist(p$age,
+   main = "Distribution of Age",
+   cex.main=1.2,
+   col.main="Black",
+   font.main=7,
+   xlab = "Age of the Retired People",
+   cex.lab=0.9,
+   font.lab=10,
+   col.lab="Black",
+   font.axis=10,
+   cex.axis=1.1,
+   col.axis="Black",
+   angle = 90,
+   col="yellowgreen",
+   xlim = c(50,75),
+   ylim = c(0,100),
+   breaks = 4,
+ )
> #2b. Graph of Quant. Variable (educ)
> min(p$educ)
[1] 8
> max(p$educ)

```

```

[1] 18
> edu=Quant.function(p$educ)
> h=hist(p$educ,
+   main = "Education Distribution",
+   cex.main=1.1,
+   col.main="Black",
+   font.main=7,
+   xlab = "Education of Retired People",
+   cex.lab=1.1,
+   font.lab=10,
+   col.lab="Black",
+   font.axis=10,
+   cex.axis=1.2,
+   col.axis="Black",
+   angle = 90,
+   col="seagreen",
+   xlim = c(8,18),
+   ylim = c(0,100),
+   breaks = 4,
+ )

```

Part 3 here:

```

> #Question 03:
> summary(p$educ)
  Min. 1st Qu.  Median   Mean 3rd Qu.   Max.
  8.00  12.00  12.00  13.57  16.00  18.00
> sd(p$educ)
[1] 2.578799
> summary(p$age)
  Min. 1st Qu.  Median   Mean 3rd Qu.   Max.
 54.00  57.00  60.00  60.48  64.00  73.00
> sd(p$age)
[1] 4.208586
> #Measures of Location for Quantative Variables (educ and age):
> #Part One (A) : Measures of Location for age:
> mean(p$age)
[1] 60.48454
> median(p$age)
[1] 60
> Modes <- function(x) {
+   ux <- unique(x)
+   tab <- tabulate(match(x, ux))
+   ux[tab == max(tab)]
+ }

```

```

> Modes(p$age)
[1] 55
> quantile(p$age)
 0% 25% 50% 75% 100%
 54 57 60 64 73

> #Part One (B): Measures of Variation:
> var(p$age)
[1] 17.71219
> sd(p$age)
[1] 4.208586
> CV=sd(p$age)/mean(p$age)
> CV
[1] 0.06958119
> range(p$age)
[1] 54 73
> min(p$age)
[1] 54
> max(p$age)
[1] 73
> Range=max(p$age)-min(p$age)
> Range
[1] 19
> quantile(p$age)
 0% 25% 50% 75% 100%
 54 57 60 64 73
> mad(p$age, na.rm = FALSE, center=median(p$age))
[1] 4.4478
> IQR(p$age)
[1] 7
> #Part One (C): Measures of Distribution:
> b=plot(density(p$age))
> #Part Two (A): ##Measures of location of educ (Education)
> mean(p$educ)
[1] 13.57216
> median(p$educ)
[1] 12
> Modes <- function(x) {
+   ux <- unique(x)
+   tab <- tabulate(match(x, ux))
+   ux[tab == max(tab)]
+ }
> Modes(p$educ)
[1] 12

```

```

> quantile(p$educ)
 0% 25% 50% 75% 100%
 8 12 12 16 18
> #Part Two (B): Measures of Variation for educ:
> var(p$educ)
[1] 6.650206
> sd(p$educ)
[1] 2.578799
> CV=sd(p$educ)/median(p$educ)
> CV
[1] 0.2148999
> min(p$educ)
[1] 8
> max(p$educ)
[1] 18
> range=max(p$educ)-min(p$educ)
> range
[1] 10
> mad(p$educ)
[1] 1.4826
> #Part Two (C): Measures of Distribution for education:
> t=plot(density(p$educ))

```

#Question 04:

#Part One: The relationship between educ and age (two quant):

```
c=cor(p$age, p$educ)
```

```
c
```

```
-0.04095366
```

```
x=cov(p$age, p$educ)
```

```
x
```

```
-0.4444741
```

#There is a weak association between age and Education of individuals whose financial decisions in retirement savings is being studied.

```
plot(p$age, p$educ, main="Scatter Plot of Education and Age", col=c("red"), xlab="Age in Years", ylab="Education in Years")
```

#The graph clearly indicates the weak relationship between the Age and Education of surveyed individuals.

#Part Two: The relationship between Gender and age (one qual one quant):

```
plot(density(p$age[p$Gender=="Male"]),
```

```
col="red",
```

```
main = "Joint Density of the Retired people based on thier Gender",
```

```
xlab = "Years of Education",
```

```
ylim = c(0.0, 0.6),  
col.lab="darkgreen",  
col.main="red",  
font.main=7,  
cex.lab=1.2,  
font.lab=6)
```

```
lines(density(p$age[p$Gender=="Male"]), col="darkgreen")
```

```
lines(density(p$age[p$Gender=="Male"]), col="red")
```

```
summary(p$age[p$Gender=="Female"])
```

```
summary(p$age[p$Gender=="Male"])
```

```
t.test(p$age~p$Gender)
```

```
t = -4.9505
```

#As long as the t.test is less than -2, therefore, there is a strong association between the two variables of Gender and Education. The years of education is negatively related to Gender of Participants.

#Part Three: The relationship between Gender and Martial Status (two qual variables):

```
chisq.test(p$Gender, p$`Martial Status`)
```

```
X-squared = 38.662
```

#Therefore, the association between the two qual variables (Gender and Martial Status) is strong.